CDB updates for v6.9

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- New calibrations
  - calEventStuffV2
  - calTileTimeCalibration
- New global tags
- Documentation

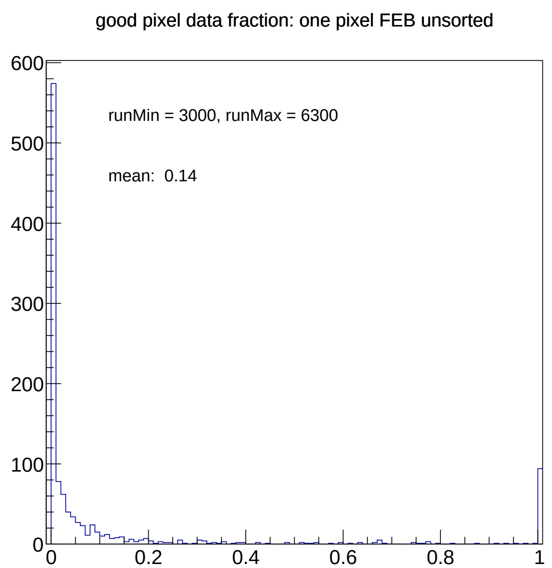
Introduction

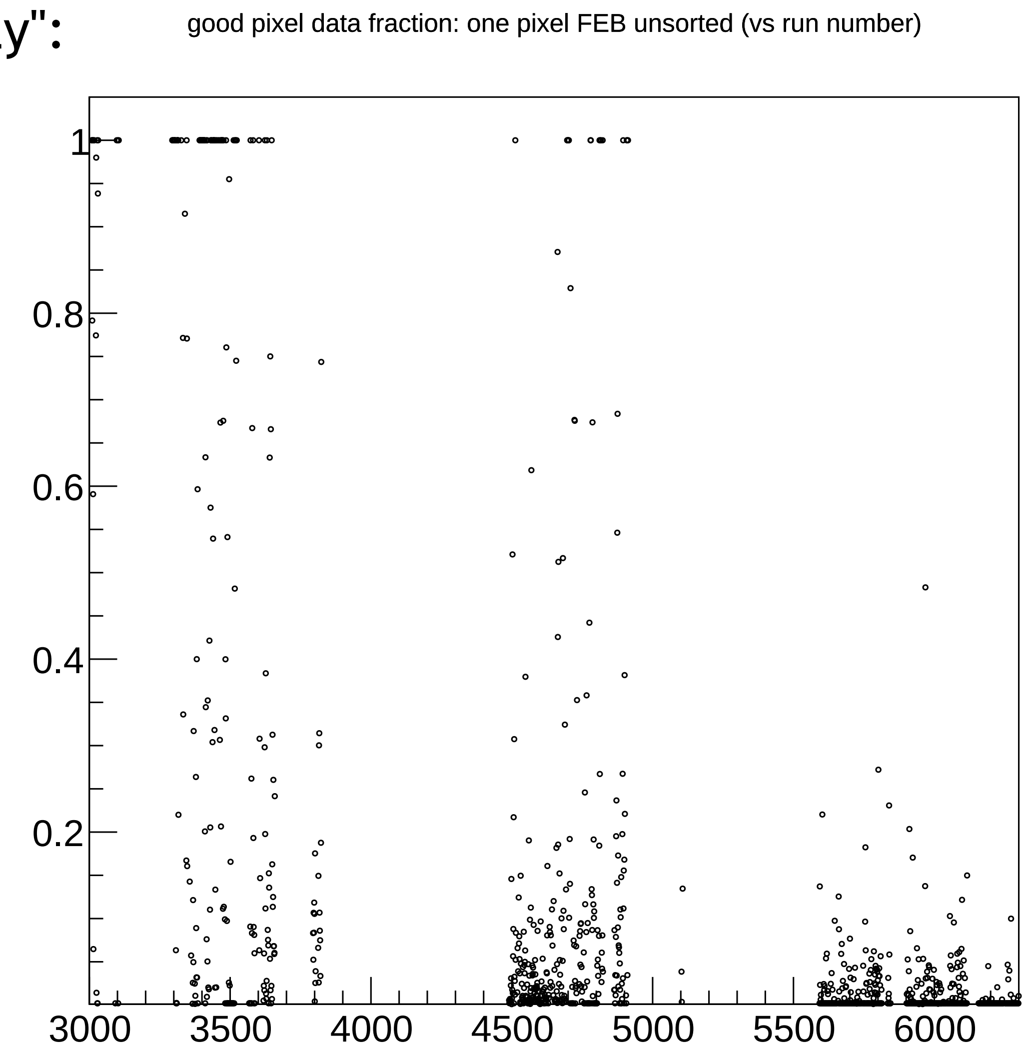
- New calibration classes
  - ▷ calEventStuffV2 add information about "skipped header bug"
  - ▷ calTileTimeCalibration first draft based on discussions with Erik
  - ▷ v6.9 does not work with old GTs - is this a problem?
    (in principle calEventStuffV2 is strictly "fyi" only)
- New payloads
  - ▷ complete run 2025 payloads for calEventStuffV2
  - ▷ single run (3274) payload for calTileTimeCalibration
- New GT - global tags
  - ▷ for v6.9
- Note
  - ▷ v6.9 GT do not work for old releases - not negotiable
  - ▷ v6.9 does not work with old GTs - speak up if you have a use case
    (to state the obvious: most old tags continue in v6.9 GTs)

calEventStuffV2 (I)

- Based on Marius' mu3e_midas_meta from mu3eUtil PR 49
  - ▷ "add check for skipped header bug"
  - ▷ affects most data "completely":
  - ▷ first_frame_at_least_one_pixel_FEB_had_unsorted_hit_data
  - ▷ varies from run to run!

⇒ Per-run data
  - ▷ IOV concept abuse
  - ▷ must check validity!
    - no payload from previous run





calEventStuffV2 (II)

- This data is strictly "for information"
  - without correction it amounts to throwing away 86% of 2025 data
  - should be correctible - might be corrected . . .
- Payload data and access

```
// -- For this payload, the only IOVs that make sense are ONE PAYLOAD PER RUN (because the
//    payload contain per-run information, e.g. EventData:endFrameData)
//
// -- IMPORTANT: Before using this payload make sure that it is valid for the run you are looking at!
bool        isValid(int run) override {return run == fIovRun;}

// -- direct accessors
uint64_t startFrameEventData() {return fConstants.eventData.startFrameData;}
uint64_t endFrameEventData() {return fConstants.eventData.endFrameData;}
uint64_t firstFrameWithFEBProblems() {return fConstants.eventData.firstFrameWithFEBProblems;}

// -- local and private
struct constants {
  struct EventData {
    uint64_t startFrameData{0};
    uint64_t endFrameData{0xffffffffffffffff};
    uint64_t firstFrameWithFEBProblems{0xffffffffffffffff};
  } eventData;
  struct PixelData { /*ditto*/ } pixelData;
  struct TileData { /*ditto*/ } tileData;
  struct FibreData { /*ditto*/ } fibreData;
```

(this information should belong into the RDB, but that is maybe too much)

calTileTimeCalibration

- Discussion with Erik
  - first version, might change
  - no "real"/validated payload data so far
- Provides calibration constants for correcting
  - differential non linearities ("dnl_corrected_time_fraction")
  - time alignment
  - time walk (energy dependent)
- Schema hopefully sufficiently flexible
  - "ui_id,d_dnl[32],d_ta,i_tw_nbins[,d_tw_ns][,d_tw_energy]"
  - i.e. binning for energy-dependent time walk can be changed

v6.9 Global tags

- datav6.9=2025V0 mcrealisticv6.9=2025V0 mcidealv6.9

```
moor>./bin/cdbSummaryGT -u ~/data/mu3e/cdb/ -g datav6.9=2025V0
cdbJSON::readGlobalTags() from  gtdir = /Users/ursl/data/mu3e/cdb//globaltags
GT: datav6.9=2025V0
  tags: 13
  comment: July 2026. Requires at least CDB-v6.9pre for mu3eUtil. Covers entire 2025 datataking period with two-layer VTX
        and installed fibres/mppcs (thanks to CX) and installed tiles. VTX alignment based on metrology and cosmics data
        (DF and MS, respectively). eventStuffV2 contains skipped header bug information. pixelQualityLM payloads
        correctly include LVDS errors from MIDAS meta data file and includes the pixel mask file information (except for
        cosmic runs >= 6865, see pixelmask tag comment). pixelEfficiency with reduced iov list: 1 (perfect), 5700 (TCS,
        still to be corrected). eventStuffV2 (with information on skipped header bug) payloads complete (except for
        crashing jobs). tileTimingCalibration payloads are placeholders so far.
```

```
 1) detsetupv1_datav6.3=2025V0
    IOV length: 3
    comment: Database entry inserted

 2) eventstuffv2_datav6.9=2025V0
    IOV length: 6035

 3) fibrealignment_mcidealv6.9=2025
    IOV length: 1
    size: 754 detector units

 4) fibrequality_datav6.3=2025V0
    IOV length: 99

 5) mppcalignment_mcidealv6.9=2025
    IOV length: 1
    size: 4 detector units

 6) pixelalignment_datav6.3=2025V1
    IOV length: 1
    size: 108 detector units
    comment: VTX only, based on metrology and
             cosmics data (DF and MS, respectively)

 7) pixelefficiency_datav6.5=2025V0
    IOV length: 2
    size: 108 detector units    IOV     5700  size matches: OK
    comment: perfect efficiencies starting from run 1 (no experimental
             numbers available), from run 5700 onwards numbers from TCS
             (likely underestimating pixel efficiency, with known bias).

 8) pixelmask_datav6.6=2025V0
    IOV length: 3
    IOV     5592  size matches: OK
    comment: pixel masks: Tune 1 and Tune 2. No proj
             available for cosmic runs (additional r
             rows/cols around edges required - DIY)

 9) pixelqualitylm_datav6.5=2025V0
    IOV length: 3330
    detector units: 108
    IOV     9361  size matches: OK
    comment: Correctly includes the LVDS errors fro
             midas meta data file

10) pixeltimecalibration_datav6.3=2025V0
    IOV length: 1
    detector units: 108

11) tilealignment_mcidealv6.9=2025
    IOV length: 1
    detector units: 1248

12) tiletimecalibration_datav6.9=2025V0
    IOV length: 2
    detector units: 1248
    IOV     3274

13) tilequality_datav6.3=2025V0
    IOV length: 13
    detector units: 5824
```

Documentation

- Reminder: mu3eUtil/cdb/test/cdbSummaryGT (with help text)

```
login002>cdbSummaryGT -u /data/project/mu3e/cdb/
cdbJSON::readGlobalTags() from  gtdir = /data/project/mu3e/cdb//globaltags
Available GTs:

datav6.1=2025Beam

datav6.3=2025V0
    comment: Do not use! November 2025. Covers entire 2025 datataking
             period with two-layer VTX and fragmentary fibres and tiles (beam and
             cosmics). Alignment from MC. pixelQualityLM payloads are incorrect!

datav6.3=2025V1
    comment: December 2025. Covers entire 2025 datataking period with
             two-layer VTX and fragmentary fibres and tiles (beam and cosmics,
             with proper B field). VTX alignment based on metrology and cosmics
             data (DF and MS, respectively).  pixelQualityLM payloads are
             incorrect!
 .. snip ..

datav6.9=2025V0
    comment: July 2026. Requires at least CDB-v6.9pre for mu3eUtil.
             Covers entire 2025 datataking period with two-layer VTX and
             installed fibres/mppcs (thanks to CX) and installed tiles.  VTX
             alignment based on metrology and cosmics data (DF and MS,
             respectively). eventStuffV2 contains skipped header bug
             information. pixelQualityLM payloads correctly include LVDS errors
             from MIDAS meta data file and includes the pixel mask file
             information (except for cosmic runs >= 6865, see pixelmask tag
             comment). pixelEfficiency with reduced iov list: 1 (perfect), 5700
             (TCS, still to be corrected).  eventStuffV2 (with information on
             skipped header bug) payloads complete (except for crashing jobs).
             tileTimingCalibration payloads are placeholders so far.
.. snip ..

mcrealisticv6.9=2025V0
    comment: July 2026. Use this for MC analysis with s/w release
             v6.9(pre). Ideal EventStuffV2. Idealized tile time calibration (no
             smearing). Pixel mask information in combination with pixelQuality
             corresponding to realistic 2025 detector conditions with
             *placeholder* MC (no) smearing for alignment/timewalk. Ideal 2025
             geometries using MC truth from v6.8 (VTX + installed fibres/mppcs) +
             installed tiles.
```

- Mu3e note under construction: Mu3e-Note-CDB_RDB

Tile geometry?

- can make corrected payloads in case this makes it into v6.9
  - ▷ mu3e PR 388 ("[sim] mv tile module 0 to top (M_PI/2)")
  - ▷ should this be backported to older GTs?

Conclusion

- Two new calibration classes with (preliminary) payloads
  - ▷ eventStuffV2
  - ▷ tileTimeCalibration
- New GTs for v6.9
  - ▷ deployed "everywhere" (mu3edb0.psi.ch/cdb, merlin6, merlin7, mu3ebe)
- new mu3eUtil PR 52 for CBD in v6.9
- Documentation improvements
  - ▷ mu3eUtil/cdb/test/cdbSummaryGT
  - ▷ Mu3e-Note-CDB_RDB
- Outlook
  - ▷ RDB API for access/upload to
    - complete run metadata information
    - complete (sub)detector DQM information
    - allowing filtering and selection
  - ▷ finish Mu3e Note 133